In this paper, without the explain, $\|\cdot\|$ means $\|\cdot\|_2$, the 2-norm.

$$A = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ A_{n1} & \cdots & \cdots & A_{nn} \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} B_{11} & B_{12} & \cdots & B_{1n} \\ B_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ B_{n1} & \cdots & \cdots & B_{nn} \end{pmatrix}$$

Denote $e_j \in F^{n\times 1}$, then

$$Ae_j = \begin{pmatrix} A_{1j} \\ A_{2j} \\ \vdots \\ A_{nj} \end{pmatrix} \quad \text{and} \quad e_i^T A = \begin{pmatrix} A_{i1} & A_{i2} & \cdots & A_{in} \end{pmatrix}$$

$$AB = \left(\sum_{k=1}^{n} A_{ik} \cdot B_{kj} \right)_{i,j} = \left(f(A^T e_i, Ae_j) \right)_{i,j},$$

where $f : F^n \times F^n \to F : \left(\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \right) \mapsto \sum_{k=1}^{n} x_k y_k$, which is the bilinear form on F^n.

$$\underbrace{\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}}_{n\times 1} \cdot \underbrace{\begin{bmatrix} b_1 & b_2 & \cdots & b_n \end{bmatrix}}_{1\times n} = \begin{bmatrix} a_1b_1 & a_1b_2 & \cdots & a_1b_n \\ a_2b_1 & \ddots & & \vdots \\ \vdots & & a_ib_j & \vdots \\ & & \ddots & \\ a_nb_1 & \cdots & \cdots & a_nb_n \end{bmatrix}$$

「Basic Matrix

「Def. Let $A \in M_{m\times n}(F)$ be given where $F=\mathbb{R}$ or $\mathbb{C}$. Define the supremum norm of A

$$\|A\|_p \overset{\text{def}}{=} \sup_{x\neq 0} \frac{\|Ax\|_p}{\|x\|_p} \overset{(*)}{=} \sup_{\|x\|_p=1} \|Ax\|_p$$

where $\|\cdot\|_p$ is the P-norm, that is,

$$\|\cdot\|_p : F^r \to \mathbb{R} : (a_i)_{i=1}^n \mapsto \left(\sum_{i=1}^n |a_i|^p\right)^{\frac{1}{p}}.$$

The equality $(*)$ is given by the fact that: for any non-zero vector x, then

$$\frac{\|Ax\|_p}{\|x\|_p} = \left\|A\cdot \frac{x}{\|x\|_p}\right\|_p = \|Ax'\|_p, \text{ where } \|x'\|=1.$$

「Prop. Given $A\in M_{m\times n}(F)$, the following conditions are satisfied:

1) $\|A\|_1 = \max_{1\le j\le n} \|Ae_j\|_1 = \max_{1\le j\le n} \sum_{i=1}^m |A_{ij}|$

2) $\|A\|_\infty = \max_{1\le j\le m} \|e_j^T A\|_1 = \max_{1\le i\le m} \sum_{j=1}^n |A_{ij}|$

3) $\|A\|_2 = \sigma_{max}(A)$, the largest singular value for A.

Proof. By definition, $\|A\|_p = \sup_{\|x\|_p=1} \|Ax\|_p$ and given $x\in F^n$ s.t $\|x\|_1=1$, $(x=(x_1,\cdots,x_n))$

$$\|Ax\|_1 = \left\|\sum_{k=1}^n x_k Ae_k\right\|_1 \le \sum_{k=1}^n |x_k|\cdot \|Ae_k\|_1 \le \sum_{k=1}^n |x_k|\cdot \max_{1\le j\le n}\|Ae_j\|_1 = \max_{1\le j\le n} \|Ae_j\|_1$$

And, take $x=e_j$ where the index $1\le j\le n$ satisfying $\|Ae_j\|$ is the maximal value, we obtain the equality.」1) And, to show the second assertion, observe that

$$\|Ax\|_\infty = \left\|\sum_{k=1}^n x_k Ae_k\right\|_\infty \le \sum_{k=1}^n |x_k|\cdot\|Ae_k\|_\infty$$

Differentiation.

Def. Let $A \subseteq \mathbb{R}^n$; let $f: A \to \mathbb{R}^m$. For $a \in \operatorname{int} A$ and $u \in \mathbb{R}^n$ with $u \neq 0$, the limit

$$f'(a;u) \overset{\text{def}}{=} \lim_{t \to 0} \frac{f(a+tu) - f(a)}{t} \quad (\text{if the limit exists})$$

is called the <u>directional derivative</u> of f at a with respect to u.

And, f is said to be differentiable at a if; there is a linear map $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ s.t.

$$\frac{f(a+h) - f(a) - T(h)}{\|h\|} \to 0 \quad \text{as } h \to 0. \ (h \in \mathbb{R}^n)$$

Such a linear map T is unique, and it is denoted $Df_a = T$, called derivative.

Thm. Let $A \subseteq \mathbb{R}^m$; let $f: A \to \mathbb{R}^n$.
If f is differentiable at a, then all the directional derivatives of f at a exist

satisfying $f'(a;u) = Df(a) \cdot u$ for all $u \in \mathbb{R}^m$.

Proof By definition, given $u \in \mathbb{R}^n$ with $u \neq 0$,

$$\lim_{t \to 0} \frac{f(a+tu) - f(a) - Df(a) \cdot (tu)}{\|t \cdot u\|} = 0.$$

Meanwhile, since $Df(tu)$ is linear, $Df(a) \cdot (tu) = t \cdot Df(a) \cdot u$.

$$\lim_{t \to 0} \left\| \frac{f(a+tu) - f(a)}{|t| \cdot \|u\|} - \frac{t \cdot Df(a) \cdot u}{|t| \cdot \|u\|} \right\|$$

$$= \lim_{t \to 0} \left\| \frac{f(a+tu) - f(a)}{t} - Df(a) \cdot u \right\| \frac{t}{|t|} \cdot \|u\| = 0$$

So proved. ▨

But, the converse is not true: observe

$$f: \mathbb{R}^2 \to \mathbb{R}; (x,y) \mapsto \frac{x^2 y}{x^4 + y^2} \text{ if } (x,y) \neq (0,0), \text{ but } (0,0).$$

Def. Given $f: \mathbb{R}^n \to \mathbb{R}^m$, define the jth partial derivative at a as

$$D_j f(a) \overset{\text{def}}{=} f'(a, e_j).$$

Thm. Given $f: \mathbb{R}^n \to \mathbb{R}^m$, denote $f_i = \pi \circ f : \mathbb{R}^n \to \mathbb{R}$. $(1 \le i \le m)$
If f_j is differentiable at a, then

$$Df_i(a) = \begin{bmatrix} D_1 f_i(a) & D_2 f_i(a) & \cdots & D_n f_i(a) \end{bmatrix}$$

Moreover,

f is differentiable at a if and only if for each f_i is differentiable at a.

Proof. The first assertion is directly, by

$$Df_i(a) \cdot e_j \underset{\text{Lemma}}{=} f_i(a; e_j) \underset{\text{Def.}}{=} D_j f_i(a)$$

And, observe that: let $T: \mathbb{R}^n \to \mathbb{R}^m$ be any linear map. Consider a map

$$F(h) = \frac{f(a+h) - f(a) - T \cdot h}{\|h\|} \quad \text{for small enough } h \in \mathbb{R}^n.$$

Since the projection is linear,

$$\pi_i \circ F(h) = \frac{f_i(a+h) - f_i(a) - \pi_i \circ T \cdot h}{\|h\|}$$

By the topological property, the limit $\lim_{h \to 0} F(h)$ exists iff $\pi_i \circ F(h) \to 0$ for each i,
Thus the first assertion gives that the result. Visually,

$$Df(a) = \left(D_j f_i(a)\right)_{ij} = \begin{bmatrix} D_1 f_1(a) & D_2 f_1(a) & \cdots & D_n f_1(a) \\ D_1 f_2(a) & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ D_1 f_m(a) & \cdots & \cdots & D_n f_m(a) \end{bmatrix}$$

$$\left(= \frac{\partial f_i}{\partial x_j}(a)\right)$$

Thm. Let A be an open subset in $\mathbb{R}^n$, and let $f : A(\subseteq \mathbb{R}^n) \to \mathbb{R}^3$.
Suppose that for all $1 \le i \le m$, $1 \le j \le n$, the partial derivatives $D_j f_i$ exists and continuous.
Then, f is differentiable. (Such a function is called of class C^1).

Proof. Since f is differentiable iff for each i, f_i is differentiable,
so given $1 \le i \le m$, showing f_i is differentiable under the condition is enough.
Let $a \in A$ be given. Given $h = (h_1, \cdots, h_n) \in \mathbb{R}^n$, Consider

$$P_0 = a$$
$$P_1 = a + h_1 \cdot e_1$$
$$P_2 = a + h_1 \cdot e_1 + h_2 e_2$$
$$\vdots$$
$$P_n = a + h_1 e_1 + \cdots + h_n e_n = a + h.$$

Then, by the telescoping,

$$f_i(a+h) - f_i(a) = \sum_{j=1}^{n} \left[f_i(P_j) - f_i(P_{j-1}) \right]$$

And, for fixed $1 \le j \le n$,

$$f_i(P_j) - f_i(P_{j-1}) = f_i(P_{j-1} + h_j e_j) - f_i(P_{j-1}) = D_j f_i(P_{j-1} + g_j e_j) \cdot h_j$$

where for some g_j, between 0 and h_j, by the Mean-value Theorem.
(If $h_j = 0$, still the equality valids.). So,

$$f_i(a+h) - f_i(a) = \sum_{j=1}^{n} D_j f_i(P_{j-1} + g_j e_j) \cdot h_j.$$

And, since for $B = [D_1 f_i(a) \ \cdots \ D_n f_i(a)]$,

$$\frac{f_i(a+h) - f_i(a) - Bh}{\|h\|} = \sum_{j=1}^{n} \frac{[D_j f_i(P_{j-1} + g_j e_j) - D_j f_i(a)] \cdot h_j}{\|h\|} \to 0 \text{ as } h \to 0.$$

「Chain Rule.

「Thm. Let $A \subseteq \mathbb{R}^n$ and $B \subseteq \mathbb{R}^m$ be open and $f: A \to \mathbb{R}^m$ and $g: B \to \mathbb{R}^p$ with $f[A] \subseteq B$. If f is differentiable at $a \in A$ and g is differentiable at $f(a) \in B$, then the Composite $g \circ f$ is differentiable at a with

$$D(g \circ f)(a) = Dg(b) \cdot Df(a).$$

Proof. Since B is open, Chose $\varepsilon > 0$ such that $B_\varepsilon(f(a)) \subseteq B$. And, since f is Continuous at a, there is $\delta > 0$ such that

$$f[B_\delta(a)] \subseteq B_\varepsilon(f(a))$$

In other says, given small enough $\varepsilon > 0$, there is $\delta > 0$ such that

$$\|x - a\| < \delta \Rightarrow \|f(x) - f(a)\| < \varepsilon.$$

So that, the Composition $g \circ f$ is well-defined on $B_\delta(a) \subseteq A$.」

Let $\Delta(h) = f(a+h) - f(a)$ for $\|h\| < \delta$. And, define

$$F(h) = \frac{\Delta(h) - Df(a) \cdot h}{\|h\|} \quad \text{for } 0 < \|h\| < \delta.$$

Since f is differentiable at a, by definition, $F(h) \to 0$ as $h \to 0$. So, since observing that

$$\|\Delta(h)\| = \|Df(a) \cdot h + \|h\| \cdot F(h)\| \leq \|Df(a)\| \cdot \|h\| + \|F(h)\| \cdot \|h\| = [\|Df(a)\| + \|F(h)\|] \cdot \|h\|.$$

Since $F(h)$ Converges, the quotient $\|\Delta(h)\| / \|h\|$ is bounded in $0 < \|h\| < \delta$.

Now, define G as $G(0) = 0$ and

$$G(k) = \frac{g(f(a) + k) - g(f(a)) - Dg(b) \cdot k}{\|k\|} \quad \text{for } 0 < \|k\| < \varepsilon.$$

So that for all $\|k\| < \varepsilon$,

$$g(f(a) + k) - g(f(a)) = Dg(b) \cdot k + \|k\| \cdot G(k).$$

Consequently, substituting $k = \Delta(h)$ and by the fact that $f(a+h) = f(a) + \Delta(h)$,

$$\frac{1}{\|h\|} \cdot [g(f(a+h)) - g(f(a)) - Dg(b) \cdot Df(a) \cdot h] = \frac{1}{\|h\|} \cdot [Dg(b) \cdot h$$